Wireless LANs: IEEE 802.11 (Wi-Fi) Architecture

Coverage: BSS, AP, ESS, Hidden/Exposed Terminal Problems, CSMA/CA with RTS/CTS, SIFS/PIFS/DIFS Intervals, 4-Address Frame

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1. 802.11 Architecture & Problems

• BSS with AP; ESS with DS. Hidden Terminal resolved via RTS/CTS handshake and NAV timer.

2. CSMA/CA Timing & Frame

• SIFS (ACKs/CTS) < PIFS < DIFS. 4 MAC addresses in 802.11 frame format.

■ KTU University High-Yield Exam Questions

• Q: Explain Hidden and Exposed terminal problems and CSMA/CA with RTS/CTS. (8 Marks)